

Georg von Westphalen

MedKayfab: Immunology & Pathology Edition



Patty the Pattern Recognizer

TLR4 = LPS
(Gram-Negative)



POWER: Sterile Alarm.

LORE: Recognizes PAMPs (microbes, e.g., flagellin via TLR5) and DAMPs (sterile host damages). Triggers $\text{NF-}\kappa\text{B}$ signaling to initiate pro-inflammatory cytokine production.

Nelly the Neutrophil

6-8 Hour
Blood Half-Life



POWER: Oxidative Burst.

LORE: Arrives first via IL-8. Uses NADPH oxidase to make superoxide (O_2^-). C3d patients lack this enzyme, suffering recurrent catalase-positive infections.

Mack the Macrophage

2 States
(M1 / M2)



POWER: Polarization.

LORE: M1 (IFN- γ activated) kills pathogens with ROS. M2 (IL-4/IL-13 activated) promotes healing and fibrosis. Presents antigen on MHC II to CD4 $^+$ T cells.

Denise the Dendritic Cell

Signal 1 + 2
Master



POWER: Cross-Presentation.

LORE: The most powerful APC. Can uniquely present exogenous viral antigens on MHC I to prime naive CD8 $^+$ T cells in the lymph nodes.

BLÖDSINN!

NK the Natural Killer (Innate Cytotoxicity)

CD16 / CD56
Positive



POWER: Missing Self Detection.
LORE: Kills virally infected or tumor cells that downregulate MHC I to hide from T cells. Uses perforin/granzyme. Performs ADCC via CD16.

Compy the Complement (The Cascade)

3 Pathways →
1 Convertase (C3)



POWER: The Unstoppable Cascade.
LORE: Classical, Lectin, and Alternative pathways converge on C3b (opsonization) and C5b-9 (MAC lysis). Regulated by CD55 (DA2) and CD59.

BLÖDSINN!

Anna Anaphylatoxin (C3a and C5a)

C5a = 100x
Potency of C3a



POWER: C5a Panic.
LORE: The alarm signals. C5a strongly attracts neutrophils and triggers mast cells. In sepsis, massive C3a causes simultaneous inflammation and neutrophil immunosuppression.

Interferoni the Type I Interferon (IFN- α/β)

cGAS-STING
Activation



POWER: Viral Early Warning.
LORE: Triggered by viral nucleic acids. Signals via JAK-STAT to upregulate PER (blocks viral translation) and increase MHC I expression on adjacent cells.

Infla the Inflammation (Acute Inflammation)

5 Cardinal Signs



POWER: Controlled Chaos.

LORE: Vasoconstriction (histamine) → Permeability (bradykinin) → Diapedesis (selectins/integrins). Resolves via IL-10/TGF- β or progresses to chronic fibrosis.

Granny Granuloma (Chronic Organization)

TNF- α Maintenance Required



POWER: The Walled Garden.

LORE: Built by macrophages + Th1 cells (IFN- γ) when acute clearance fails. Anti-TNF therapy (infliximab) destroys the wall, risking fatal reactivation of latent TB.

Tommy T Cell (Development & Subsets)

2 Signals Required



POWER: The Two-Signal Verification.

LORE: Signal 1 = TCR + MHC.

Signal 2 = Co-stimulation (CD28 + CD80/86).

Without Signal 2, the T cell enters anergy (permanent shutdown).

Theo Th1 (Cell-Mediated Immunity)

IL-12 Driven Differentiation



POWER: Macrophage Enragement.

LORE: Fights intracellular pathogens. Secretes IFN- γ to supercharge macrophages and upregulate MHC II. Drives delayed-type hypersensitivity and autoimmune tissue damage.

Tara Th2 (Humoral & Anti-Parasitic)

GATA-3
Transcription



POWER: Parasite Purge.

LORE: Driven by IL-4. Secretes IL-4 (IgE class switch), IL-5 (eosinophil activation), and IL-13. The primary driver of Type I hypersensitivity (allergy/asthma).

Rex Treg (The Peacekeeper)

CD4+ CD25+
FoxP3+



POWER: Civil War Prevention.

LORE: Suppresses immunity via IL-10, TGF- β , and CTLA-4 contact. FoxP3 mutation causes IPEX syndrome (lethal multi-organ autoimmunity in infant boys).

Barb B Cell (Activation & Switching)

CD40-CD40L
Binding Required



POWER: Class Switch Recombination.

LORE: AID enzyme alters heavy chains. T-cell help (CD40L) is required for memory and switching. BTK mutation causes X-linked agammaglobulinemia (no mature B cells).

Iggy Immunoglobulin (Isotypes)

5 Modular
Outfits



POWER: Fc Effector Modular Swapping.

LORE: IgM (pentamer, immediate complement activator). IgG (most abundant, crosses placenta). IgA (dimer, mucosal). IgE (mast cell bound, allergies).

CARD 17

SADIE THE CYTOTOXIC T CELL (CD8+ CTL)

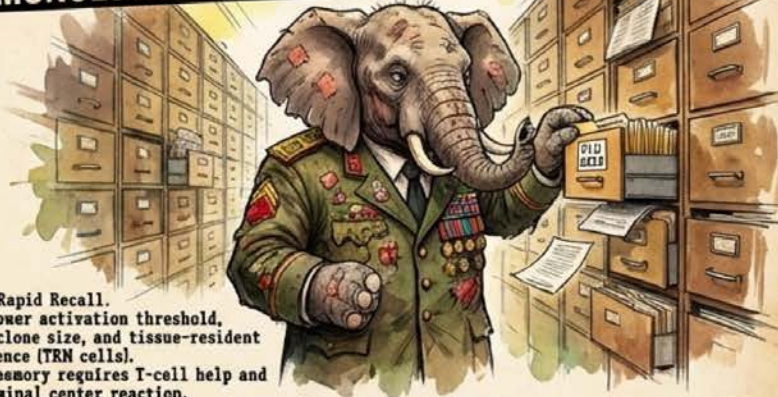
MHC I
RESTRICTED

POWER: Perferin Punch.

LORE: Recognizes antigen on MHC I.
Kills via perforin (peres) + granzymes B (speptesis) or Fas-FasL. Inhibited by PD-1/PD-L1 checkpoint exhaustion in tumors.

CARD 18

MEMO THE MEMORY CELL (IMMUNOLOGICAL MEMORY)

DAYS VS. WEEKS
KINETICS

POWER: Rapid Recall.

LORE: Lower activation threshold,
larger clone size, and tissue-resident
persistence (T_H cells).
B-cell memory requires T-cell help and
the germinal center reaction.

CARD 19

CYNDY CYTOKINE (IL-1, IL-6, TNF-α)

THE INFLAMMATORY
TRIAD

POWER: Escalation.

LORE: IL-1β (produced by NLRP3
inflammation). IL-6 (drives CRP in
liver, elevated in COVID-19).
TNF-α (drives cachexia and sepsis).
Suppressed by IL-10.

CARD 20

KYLIE CYTOKINE STORM (HYPERFERRITINEMIC SYNDROME)

EXTREME
MACROPHAGE
ACTIVATION

POWER: Hyperferritinemic Bampage.

LORE: Uncontrolled IL-6, IPB-γ, and
macrophage amplification. In primary
Bemphagocytic Lymphohistiocytosis
(HLR), perforin gene mutations prevent
CTLs from shutting off the loop.

GW

Georg v. Westphalen

Gell and Coombs (The Taxonomists)

4 Mechanisms of Mischief



POWER: The Four Mischiefs.

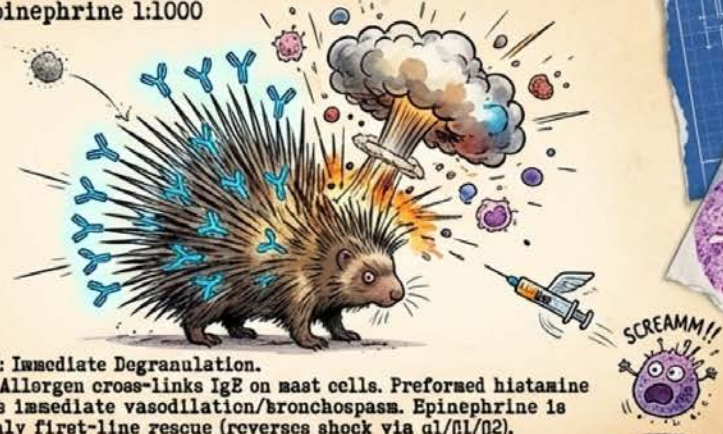
LORE: I: Immediate IgE. II: Antibody-mediated surface cytotoxicity. III: Soluble immune complexes depositing in tissue. IV: Delayed cell-mediated (48-72h).

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Ana Anaphylaxis (Type I Hypersensitivity)

IM Epinephrine 1:1000



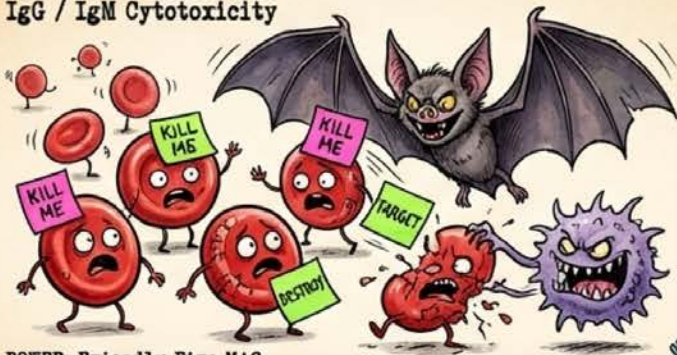
POWER: Immediate Degranulation.

LORE: Allergen cross-links IgE on mast cells. Preformed histamine causes immediate vasodilation/bronchospasm. Epinephrine is the only first-line rescue (reverses shock via $\alpha_1/\beta_1/\beta_2$).

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Hemo the Hemolytic (Type II Hypersensitivity)

IgG / IgM Cytotoxicity



POWER: Friendly Fire MAC.

LORE: Antibodies target specific cell surfaces. Causes Goodpasture's (anti-GBM), Autoimmune Hemolytic Anemia, and Myasthenia Gravis (receptor blockade without direct lysis).

Icko Immune Complex (Type III Hypersensitivity)

Soluble Ag-Ab Lattices



POWER: Glomerular Trap.

LORE: IgG complexes form in blood during antigen excess. They deposit in small vessels (glomeruli, synovium) → complement activation → neutrophil destruction (e.g., SLE, Serum Sickness).

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DELLA DELAYED (TYPE IV HYPERSENSITIVITY)

48-72 HOUR PEAK



PONER: Th1 Grudge.

LORE: Cell-mediated, no antibodies! CD4+ Th1 cells recognize antigen, secrete IFN- γ , and activate macrophages. Drives contact dermatitis, tuberculin skin test, and Type 1 Diabetes.

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MASTO THE MAST CELL (TISSUE BIOLOGY)

Fc ϵ RI PERSISTENTLY LOADED

PONER: Leukotriene Sustenance.

LORE: Activation releases preformed histamine (minutes). Newly synthesized leukotrienes (hours later) cause bronchoconstriction 1000x more potent than histamine. Blocked by montelukast.

KayfaBOGGLE!

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ALLIE ALLERGY (IgE-MEDIATED SPECTRUM)

Th2 SKEWING



POWER: The Allergic March.

LORE: Epithelial defect $\rightarrow$ Th2 dominance (IL-4/IL-5/IL-13). Allergen immunotherapy desensitizes by shifting the response from Th2 to Treg and inducing IgG4 blocking antibodies.

EOSIE THE EOSINOPHIL (ANTI-PARASITE / ALLERGY)

 $>1.5 \times 10^9/L$ in HES

POWER: Major Basic Toxicity.

LORE: Releases Major Basic Protein (MBP) to kill helminths. Driven by IL-5 (blocked therapeutically by mepolizumab). Causes tissue remodeling in asthma and Hypereosinophilic Syndrome.

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Lupi the Lupus (Systemic Lupus Erythematosus)

Anti-dsDNA / Anti-Sm



PONER: The Great Imitator.

LORE: Defective apoptotic cell clearance exposes nuclear antigens. TLR9/TLR7 activate Type I IFN and autoantibody production. Treat all patients with hydroxychloroquine.



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Rhea Rheumatoid (Rheumatoid Arthritis)

HLA-DR4 (Shared Epitope)



POWER: Pannus Invasion.

LORE: Synovial inflammation drives invasive pannus formation → cartilage/bone erosion. Anti-CCP antibodies (ACPA) are highly specific. Synovial fluid is neutrophil-rich with low glucose.

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Sid Sjögren (Primary Sjögren's Syndrome)

50x B-Cell Lymphoma Risk



PONER: Glandular Drought.

LORE: Lymphocytic infiltration destroys salivary/lacrimal glands (anti-Ro/SSA, anti-La/SSB). Causes extreme xerostomia/dry eyes, and can cause Type 1 Renal Tubular Acidosis.



BRIQENT

32

Scleroder Max (Systemic Sclerosis)

Anti-Scl-70 / Anti-Centromere



POWER: Fibrosis Architecture.

LORE: Endothelial injury → Raynaud's → TGF- β driven fibrosis. Diffuse (anti-Scl-70, early ILD) vs. Limited (anti-centromere). Renal crisis MUST be treated with ACE inhibitors despite AKI.

INFLAMMA THE MYOPATHY (INFLAMMATORY MYOPATHIES)

ELEVATED
CREATINE KINASE



POWER: CD8+ Friendly Fire.

LORE: Polymyositis: CD8+ CTLs destroy muscle expressing aberrant MHC I. Dermatomyositis: Type I IFN drives microangiopathy (Gotttron's papules, hellotrope rash). Anti-Jo-1 = antisynthetase syndrome.

BETTY B27

(SERONEGATIVE SPONDYLOARTHROPATHIES)

~90% ANKYLOSING
SPONDYLITIS

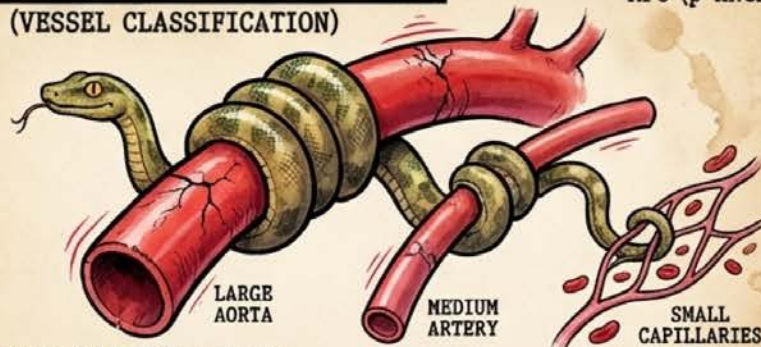


POWER: The Backdoor Spondyloarthritis.

LORE: HLA-B27 misfolding triggers the IL-23/IL-17 axis. Causes AS, Psoriatic, Reactive (can't see, pee, climb a tree), and IBD-arthritis. Sacroiliitis and enthesitis are hallmarks.

VASCY VASCULITIS (VESSEL CLASSIFICATION)

PR3 (c-ANCA) vs
MPO (p-ANCA)



POWER: Caliber Constriction.

LORE: Large (Giant Cell Arteritis → blindness risk, immediate steroids). Medium (PAN, Kawasaki). Small (ANCA-associated: GPA/Wegener's = PR3; MPA = MPO).

THYRA THYROID (HASHIMOTO'S & GRAVES')

ANTI-TPO VS TSI
ANTIBODIES



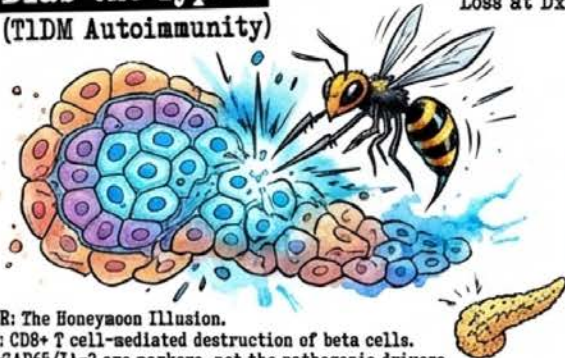
POWER: Receptor Mimicry.

LORE: Hashimoto's: Th1-mediated destruction → hypothyroidism. Graves': IgG mimics TSH, continually activating the receptor → hyperthyroidism and proptosis (orbital fibroblasts express TSH-R).

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Diab the Type 1 (T1DM Autoimmunity)

~80% Beta Cell
Loss at Dx



POWER: The Honeymoon Illusion.

LORS: CD8+ T cell-mediated destruction of beta cells.

Anti-GAD65/IA-2 are markers, not the pathogenic drivers.

Teplizumab (anti-CD3) can delay clinical onset in high-risk patients.



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Good the Goodpasture (Anti-GBM Disease)

Type IV Collagen
 $\alpha 3$ Target



POWER: Linear IP Strike.

LORS: IgG antibodies directly attack the GBM/alveolar membranes.

Causes pulmonary hemorrhage + Rapidly Progressive GN.

Immunofluorescence shows smooth, LINEAR IgG deposition.

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Scid the Severe Combined (SCID)

IL-2R γ
(X-linked)
or ADA



POWER: Complete Void.

LORS: Complete failure of T-cell development (and often B/KK cells).

Fatal if given live vaccines (e.g., BCG-osis). Maternal T-cells crossing the placenta can cause fatal GVHD in the infant.



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Aggie Agammaglobulinaemia (XLA)

BTX Gene
Mutation



POWER: Encapsulated Vulnerability.

LORS: Pro-B fails to transition to pre-B. No mature B cells, no tonsils, no germinal centers. Highly susceptible to encapsulated bacteria (pneumococcus, H. Influenzae) after maternal IgG fades at 6 months.

CODY COMMON VARIABLE (CVID)

41

Most Common
Adult PID



POWER: Late-Onset Failure.

LORE: Normal B-cell numbers but defective antibody production (low IgG/IgA). Paradoxically presents with recurrent sinopulmonary infections, severe autoimmunity (cytopenias), and highly increased lymphoma risk.

CELL WARS!

TRANS THE TRANSPLANT (Rejection & GVHD)

43

Hyperacute
= Minutes



POWER: Graft-vs-Host Betrayal

LORE: Hyperacute rejection is driven by preformed donor-specific antibodies (prevented by crossmatching). GVHD occurs when donor T-cells attack recipient tissues (skin, gut, liver).

CELL WARS!

IMMUNE SYSTEM
SABOTAGE!

COMPLE COMPLEMENT DEFICIENCY

42

C5-C9 Loss =
MAC Failure



POWER: Neisseria Target.

LORE: NAC deficiency (C5-C9) specifically predisposes to meningococcal/gonococcal infections. C1-INH deficiency causes bradykinin-driven hereditary angioedema (unresponsive to epinephrine).

CYCLO THE CYCLOSPORIN & TACROLIMUS

44

Tacrolimus
100x Potency



POWER: Calcineurin Blockade

LORE: Block cyclophilin/FKBP12 → inhibit calcineurin → block NFAT dephosphorylation → prevent IL-2 transcription. Highly nephrotoxic (afferent arteriolar constriction).

IMMUNE
SYSTEM
SABOTAGE!

RAPA the mTOR (Sirolimus)

45

G1 to S Phase
Block



POWER: G1-S Barricade.
LORE: Binds FRBP12 to inhibit mTOR. Blocks IL-2-driven T-cell proliferation (not production). Non-nephrotoxic. Impairs wound healing; used post-transplant and for LAM.

MYCO the Mycophenolate (MMF)

46

INPDH
Inhibition



POWER: De Novo Starvation.
LORE: Inhibits inosine monophosphate dehydrogenase (IMPDH). Selectively starves T and B cells, which uniquely rely entirely on de novo purine synthesis and lack a salvage pathway. Teratogenic.

ANTI-TNF the Alliance (Infliximab/Adalimumab)

47

Mandatory
Latent
TB Screen



POWER: Granuloma Breaker.
LORE: Blocks TNF- α signaling. Highly effective for RA and IBD. Because TNF maintains granulomas, blockade risks fatal reactivation of latent TB. (Etanercept is a decoy receptor, fails in IBD).

CHECK the Checkpoint (CTLA-4 & PD-1 Blockade)

48

Reverses
T-Cell
Exhaustion



POWER: Brake Release.
LORE: Anti-CTLA-4 (ipilimumab) and anti-PD-1 (pembrolizumab) remove inhibitory brakes on T-cells to fight tumors. Causes severe immune-related adverse events (irAEs) like colitis and pneumonitis.

JAK the JAK Inhibitor (Tofacitinib, etc.)

Oral, Reversible,
Rapid



POWER: Multi-Cytokine Silence

LORE: Blocks intracellular JAK-STAT signaling, silencing dozens of cytokine receptors simultaneously. Causes high risk of Varicella Zoster (shingles) reactivation—check immunity before starting.

IL-17 Ivan (Secukinumab)

Secukinumab
Worsens IBD



POWER: Epithelial Enragement

LORE: IL-17 (from Th17) drives psoriasis and ankylosing spondylitis entheitis. Anti-IL-17 cures the skin/joints but worsens Inflammatory Bowel Disease because IL-17 is protective in the GI mucosa.

CELL the Cellular Pathology (Adaptation vs Death)

4 Necrosis
Morphologies



POWER: Adaptation

LORE: Apoptosis is clean; Necrosis is inflammatory. Coagulative necrosis (ischemia) leaves ghost outlines. Liquefactive necrosis (brain/abscesses) causes enzymatic self-digestion.

NEC Necroptosis & Pyroptosis

Gasdermin D
Pores



POWER: Programmed Inflammation

LORE: Death that shouts.
Pyroptosis uses the NLRP3 inflammasome → caspase-1 → IL-1 β release via gasdermin pores. Gain-of-function NLRP3 mutations cause CAPS (treat with anakinra).

KayfaBIZARRO!



FERRI the Fibrosis (Pathological Healing)

53

TGF- β 1
Master Switch



POWER: Rogue Healing.
LORE: Persistent TGF- β signaling activates Smad2/3 $\rightarrow$ fibroblast-to-myofibroblast transition $\rightarrow$ excessive architectural collagen (e.g., IPF, Cirrhosis). Pirfenidone blocks this pathway.

ONCO the Oncogenesis (Hallmarks of Cancer)

54

p53 Mutated
in >50%

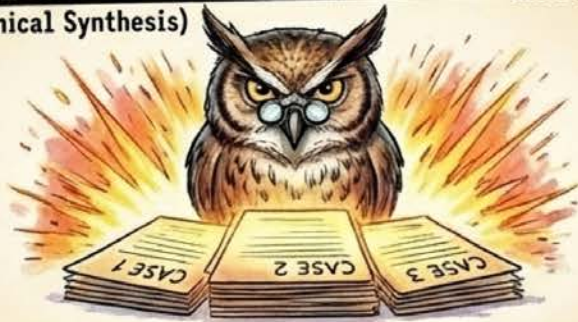


POWER: The Ten Broken Rules.
LORE: Hallmarks include immune evasion, replicative immortality, and the Warburg effect (aerobic glycolysis). Tumors actively recruit M2 macrophages and Tregs to suppress the local immune response.

THE IMMUNOLOGY ATTENDING (Clinical Synthesis)

55

Localize
the Defect



POWER: Master Clinical Synthesis.
LORE: The ultimate quest. Every drug, mechanism, and pathology card interacts. If you give Anti-TNF to a RA patient, you disable Granny Granuloma. If you lack BTK, you lose Barb B-Cell.

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FrizzleBob's Kayfabe Biologic Matrix

Biologic Class	Target	Cures	WORSENS/Risks
Anti-TNF (Infliximab)	TNF- α	RA, Crohn's, Psoriasis	Worsens MS Reactivates TB
JAK Inhibitor (Tofacitinib)	JAK-STAT	RA, UC	Shingles (VZV) Reactivation
Anti-IL-17 (Secukinumab)	IL-17A	Ankylosing Spondylitis	Triggers/Worsens IBD
Anti-IL-4R α (Dupilumab)	Th2 Axis	Asthma, Atopic Derm	Eosinophilia (transient)

KayfaBIZARRO!